MAKSUDA AKTHER TUMPA

Aspiring AI Engineer | Information Technology Undergraduate

Cyberjaya, Selangor, Malaysia | maksudaaktertumpa@gmail.com

linkedin.com/in/maksudaakhtertumpa | github.com/maksudaakhtertumpa

PROFESSIONAL SUMMARY

Final-year IT student (CGPA 3.65, Dean's List) building LLM applications with Python, FastAPI and RAG. Experienced in hybrid search, reranking, source-cited answers, retrieval evaluation and automated testing. Seeking an AI engineering internship or entry-level role.

EDUCATION

Bachelor of Information Technology, University of Cyberjaya Jul 2024 - Present

Expected graduation Jul 2027 | CGPA 3.65 | Dean's List 2025/2026

- **Coursework:** Artificial Intelligence, Data Mining and Knowledge Discovery, Data Science, Object-Oriented Programming, Database Systems, Web Application Development

PROJECTS

Enterprise Knowledge AI: Hybrid Search RAG Platform GitHub repository

Python | FastAPI | ChromaDB | SQLite FTS5 | Ollama | Next.js | pytest | Playwright

- Building a document Q&A platform on open-source models (Qwen3 embeddings, bge reranker, Qwen 3.5 via Ollama), running locally without paid API calls.
- Developed a FastAPI REST API for document upload, search and Q&A, with per-workspace API keys and data isolation.
- Benchmarked BM25, dense, hybrid (RRF) and reranked retrieval; reranking reached **MRR 0.962** and **Recall@5 1.000** vs. 0.538 MRR for BM25 (development benchmark).
- Every answer cites its source, checked word for word against retrieved text. Development tests: **29/30** response coverage, **11/12** correct refusals, and all **4** tested prompt-injection cases resisted.
- Automated tests (pytest, Vitest, Playwright) and linting run in GitHub Actions CI.

AskUoC: AI Assistant for the University of Cyberjaya GitHub repository

Python | LangGraph | FastAPI | PostgreSQL + pgvector | Gemini | Next.js | Docker

- Built a RAG chatbot over 872 university web pages and 106 PDFs (5,218 chunks) that cites a source for every answer.
- Designed a LangGraph pipeline (query rewriting, hybrid search with RRF, reranking, relevance grading) that answers "not found" instead of guessing.
- Fixed wrong fee answers by chunking fee tables one row per programme; tuned retrieval on a 28-question development test set.
- Delivered FastAPI backend, Next.js chat UI and admin console, and a CI pipeline with **182** backend tests, **19** Playwright end-to-end tests and security scans.

FRESHO Web Application (team university project)

React | Tailwind CSS

- Built responsive UI components and contributed to functionality testing and documentation.

TECHNICAL SKILLS

Programming: Python, TypeScript, JavaScript, SQL

AI / LLM: RAG, LangGraph, LangChain, embeddings, hybrid search (BM25, RRF), reranking, Ollama, Gemini API, RAG evaluation

Backend & Data: FastAPI, REST APIs, PostgreSQL, pgvector, ChromaDB, SQLite, Redis

Web & Tools: Next.js, React, Tailwind CSS, Git, GitHub Actions, Docker, pytest, Playwright, Linux

CERTIFICATIONS

- **Build AI Agents using MCP**, IBM, Coursera
- **RAG for Generative AI Applications Specialization**, IBM Skills Network, Coursera
- **Develop Generative AI Applications: Get Started**, IBM, Coursera
- **SQL for Data Science**, University of California, Davis, Coursera

LANGUAGES

Bengali (Native) | English (Full Professional) | Hindi (Conversational)